

FORMAL LANGUAGES AND AUTOMATA THEORY

QUIZ # 1

REG#: _____

NAME: _____

COURSE CODE: CS224

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TOTAL TIME: 20m

TOTAL MARKS: 17

1) Construct a regular expression for?

(5 MARKS)

A. All strings that start with double letter

$$(aa + bb)(a + b)^*$$

B. All strings that do not start with double letter.

$$(ab + ba)(a + b)^* + a + b + \Lambda$$

2) Let $S = \{ab\,bb\}$, $T = \{ab\,bb\,bbbb\}$ and $R = \{ab\,bb\,bbb\}$

(5 MARKS)

Prove that that $S^* = T^*$ but $S^* \neq R^*$

$$\underline{S^* = T^*}$$

$S^* = \{\Lambda \text{ and all possible concatenations of } ab \text{ and } bb\}$

$T^* = \{\Lambda \text{ and all possible concatenations of } ab, bb, \text{ and } bbbb\}$, but $bbbb$ is just bb concatenated with itself, so

$T^* = \{\Lambda \text{ and all concatenations of } ab \text{ and } bb\} = S^*$.

$$\underline{S^* \neq R^*}$$

$S^* = \{\Lambda \text{ and all possible concatenations of } ab \text{ and } bb\}$

$R^* = \{\Lambda \text{ and all possible concatenations of } ab, bb, \text{ and } bbb\}$

S^* cannot contain the words of like bbb , $bbbb$ (odd number of b 's), so $S^* \neq R^*$

3) Consider a simple calculator which can only take two numbers, and can only perform the following operations on them: **(7 MARKS)**

- A. Difference of two single digit numbers
- B. Product of two single digit numbers
- C. Division of two single digit numbers

Your task is to design a language for the above-mentioned calculator.

Note:

1. Only one of the three operations can be performed at a time.
2. Only two numbers are allowed at a time, i.e., any of the three operations will be performed on the same two numbers.

a) Show the alphabet, Σ , of the above calculator? (3)

$$\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, -, \times, /\}$$

b) Construct the regular expression accepting the language of the calculator? (4)

Let

$$a, b = 0 + 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9$$

$$op = - + \times /$$

$$(a - b) + (a \times b) + (a / b)$$

Note: here '+' is not a symbol belonging to the Language of calculator, rather it corresponds to the 'or' operation of regular expressions.